

Problem 1

Let $a, b \in X$. Then consider $f(x) : X \rightarrow \mathbb{R}$ s.t. $f(x) = d(a, x)$. This is continuous since d is continuous.

Since X is connected we know that $f(a) = 0, f(b) = d(a, b) \in f(X)$ with $0 < d(a, b)$ this means that $[0, d(a, b)] \subseteq f(X)$.

However, in $\mathbb{R}$ this interval has an uncountable number of elements. This means that X is uncountable. Since if X were countable then this would imply $f(X)$ is countable which it is not thus X is uncountable. Meaning it has an uncountable number of points.'

Problem 2

Let $x, y \in \mathbb{R}^\omega$ consider $f : [0, 1] \rightarrow \mathbb{R}^\omega$ s.t.

$$f(t) = (1 - t)x + ty$$

This is continuous path since $f(t) = ((1 - t)x_1 + ty_1, (1 - t)x_2 + ty_2 + \dots)$ we know that each coordinate is continuous since it is made up of polynomials thus f is continuous with $f(0) = x$ and $f(1) = y$.

Thus, $\mathbb{R}^\omega$ is path connected.

Problem 3

(a) Consider:

$$A = \{0\} \cup \{1/n \mid n \in \mathbb{N}\}$$

Let $x \in \mathbb{R} \setminus A$.

If $x < 0$:

Then fix $\epsilon = |x|/2 > 0$ thus $x + \epsilon < 0$ thus $B(x, \epsilon) \cap A = \emptyset$.

If $x > 1$:

Then fix $\epsilon = |x - 1|/2 > 0$ thus $x - \epsilon > 1$ thus $B(x, \epsilon) \cap A = \emptyset$

If $x \in (0, 1) \setminus A$:

Then there is an $n \in \mathbb{N}$ s.t.

$$\frac{1}{n+1} < x < \frac{1}{n}$$

Fix $\epsilon = 1/n \min(x - \frac{1}{n+1}, \frac{1}{n} - x)$

Thus,

$$\frac{1}{n+1} < x - \epsilon < x < x + \epsilon < \frac{1}{n}$$

$$B(x, \epsilon) \cap A = \emptyset$$

Thus, $\mathbb{R} \setminus A$ is open thus A is closed. Since A is bounded by Heine-Borel A is compact in $\mathbb{R}$.

Consider $\{1/n\}$ for $n \in \mathbb{N}$ this is trivially connected.

Fix $\epsilon > 0$ small enough s.t. $1/(n+1) < 1/n - \epsilon < 1/n + \epsilon < 1/(n-1)$ thus $(1/n - \epsilon, 1/n + \epsilon) \cap A = \{1/n\}$ thus $\{1/n\}$ is open.

Also, $((-\infty, 1/n) \cup (1/n, \infty)) \cap A = A \setminus \{1/n\}$ thus $\{1/n\}$ is also closed.

Assume for contradiction there is a connected subset C of A s.t. $1/n, a \in C$ s.t. $a \neq 1/n$.

Thus, $C = \{1/n\} \cup A \setminus C$ which contradicts the fact that C is connected.

Thus, $\{1/n\}$ is a connected component. Therefore, A is compact and has infinite connected components.

(b) Assume X, Y are path connected spaces. Consider $(x, y), (a, b) \in X \times Y$.

Since X is path connected there is a continuous function $f : [c, d] \rightarrow X$ s.t. $f(c) = x$ and $f(d) = a$.

Since Y is path connected there is a continuous function $g : [s, t] \rightarrow Y$ s.t. $g(s) = y$ and $g(t) = b$

Consider $\varphi : [0,1] \rightarrow [c,d]$ s.t. $\varphi(t) = c + t(d - c)$.

This is a continuous function thus $f \circ \varphi$ is continuous. Also $f \circ \varphi(0) = f(c) = a$ and $f \circ \varphi(1) = f(d) = a$

Do something similar for g . Call these new functions $\bar{g}, \bar{f}$.

Consider $h : [0,1] \rightarrow X \times Y$ s.t. $h(t) = (\bar{g}, \bar{f})$. h is continuous since $\bar{g}$ and $\bar{f}$ are continuous.

Thus, $h(0) = (\bar{g}(0), \bar{f}(0)) = (x, y)$ and $h(1) = (\bar{g}(1), \bar{f}(1)) = (a, b)$

Therefore, $X \times Y$ is path connected.

(c) Assume X is compact, and $\sim$ is an equivalence relation on X .

Let q be the quotient function.

Let $\{U_i\}_{i \in J}$ be an open cover of $X/\sim$. This means that $\{q^{-1}(U_i)\}_{i \in J}$ is an open cover of X since q is continuous thus $q^{-1}(U_i)$ is open.

Since X is compact there is a finite subcover. Thus $X = q^{-1}(U_1) \cup \dots \cup q^{-1}(U_n)$. Since q is surjective $q(q^{-1}) = U_1$. Thus $q(X) = X/\sim = U_1 \cup \dots \cup U_n$.

Therefore, $X/\sim$ is compact.

Problem 4

Let A, B be disjoint compact subsets of a Hausdorff space X . For every $a \in A$ and $b \in B$, since X is Hausdorff, there exist open sets $U_{a,b}$ and $V_{a,b}$ such that

$$a \in U_{a,b}, \quad b \in V_{a,b}, \quad \text{and} \quad U_{a,b} \cap V_{a,b} = \emptyset.$$

Now, fix $a \in A$ and consider the collection

$$\{V_{a,b}\}_{b \in B}.$$

This forms an open cover of B . Since B is compact, there exists a finite subcover

$$V_{a,b_1}, V_{a,b_2}, \dots, V_{a,b_n}.$$

Define

$$V_a = V_{a,b_1} \cup V_{a,b_2} \cup \dots \cup V_{a,b_n},$$

which is open. Also define

$$U_a = U_{a,b_1} \cap U_{a,b_2} \cap \dots \cap U_{a,b_n},$$

which is also open.

We claim that

$$U_a \cap V_a = \emptyset.$$

To see this, assume for contradiction that there exists $x \in U_a \cap V_a$. Then $x \in U_a$ and $x \in V_a$. Hence, there is some i (with $1 \leq i \leq n$) such that $x \in V_{a,b_i}$ and $x \in U_{a,b_i}$. But this contradicts the fact that

$$U_{a,b_i} \cap V_{a,b_i} = \emptyset.$$

Thus, we must have $U_a \cap V_a = \emptyset$.

Since $\{U_a\}_{a \in A}$ is an open cover of A and A is compact, there exists a finite subcover

$$U_{a_1}, U_{a_2}, \dots, U_{a_m}.$$

Now, define

$$U = U_{a_1} \cup U_{a_2} \cup \dots \cup U_{a_m}$$

and

$$V = V_{a_1} \cap V_{a_2} \cap \dots \cap V_{a_m}.$$

Since each V_{a_i} is a union of finitely many open sets that cover B , we have

$$B \subseteq V.$$

Similarly, since U is a union of open sets covering A , it follows that

$$A \subseteq U.$$

Assume for contradiction that there exists $x \in U \cap V$. Then, $x \in U_{a_i}$ for some i (with $1 \leq i \leq m$) and $x \in V_{a_i}$. This contradicts the established fact that

$$U_{a_i} \cap V_{a_i} = \emptyset.$$

Thus, $U \cap V = \emptyset$.

In conclusion, there exist disjoint open sets U and V in X such that

$$A \subseteq U \quad \text{and} \quad B \subseteq V.$$

This completes the proof.

Problem 5

1. Consider $f : \mathbb{R} \rightarrow \{0,1\}$ where $\{0,1\}$ has the discrete topology. This is an open map since everything is open in the codomain.

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ 1 & \text{if } x > 0. \end{cases}$$

Consider $f^{-1}(0) = (-\infty, 0]$ this is closed thus f is not continuous.

2. Assume $f : X \rightarrow Y$ is a surjective continuous open map.

Let $U \subseteq Y$ be open then $f^{-1}(U)$ is open by definition of continuous.

Let $U \subseteq Y$ s.t. $f^{-1}(U)$ is open. Since f is surjective $f(f^{-1}(U)) = U$. Also since it is an open map this implies that U is open.

Thus, f is a quotient map.

3. Consider $\pi|_A : A \rightarrow \mathbb{R}$

$\pi|_A$ is surjective since for $x \in \mathbb{R}$ consider $(x, 0) \in A$ thus $\pi|_A(x, 0) = x$

Let $U \subseteq \mathbb{R}$ be open then $(\pi|_A)^{-1}(U) = \pi^{-1}(U) \cap A$ this is open in A since π is continuous thus $\pi^{-1}(U)$ is open in X .

Let $U \subseteq \mathbb{R}$ s.t. $(\pi|_A)^{-1}(U)$ is open in A . Let $a \in U$, and fix $b \in \mathbb{R}$ s.t. $\pi(a, b) = a$ and $(a, b) \in A$.

Since $(\pi|_A)^{-1}(U)$ is open there is an $\epsilon > 0$ s.t. $B((a, b), \epsilon) \cap A \subseteq (\pi|_A)^{-1}(U)$.

If $a < 0$ then we know that $b = 0$.

Fix $\delta = \min(|a|/2, \epsilon/2)$

Consider $B(a, \delta) \subseteq \mathbb{R}$. Let $x \in B(a, \delta)$ then $x < 0$

thus $(x, 0) \in B((a, b), \epsilon) \cap A \subseteq (\pi|_A)^{-1}(U)$. Therefore, $x \in U$ since $\pi|_A((\pi|_A)^{-1}(U)) = U$ since $\pi|_A$ is surjective.

If $a \geq 0$ then $b \in \mathbb{R}$

If $a > 0$ fix $\delta = \min(|a|/2, \epsilon/2)$

If $a = 0$ fix $\delta = \epsilon/2$

If $b > 0$ fix $\alpha = \min(|a|/2, \epsilon/2)$

If $b = 0$ fix $\alpha = \epsilon/2$

Consider $B(a, \delta) \times B(b, \alpha) \subseteq B((a, b), \epsilon)$

Let $x \in B(a, \delta)$ and $y \in B(b, \alpha)$ since $x \geq 0$ this implies that $(x, y) \in A$ thus $(x, y) \in B((a, b), \epsilon) \cap A$. Thus, $x \in U$ for same reason as above.

Thus, in all cases we can find an open ball around $a \in U$ s.t. it is a subset of U . Thus, U is open.

Therefore, $\pi|_A$ is a quotient map.

Consider $U = A \cap \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$ which is open in A .

$\pi|_A(U) = [0, \infty)$ thus $\pi|_A$ is not an open map.

Problem 6

- (a) First since $\|x\|_{Euc}$ is made up of additions and multiplications and non-negative square roots we know that it is continuous.

Let $\{e_1, \dots, e_n\}$ be the standard basis for $\{e_1, \dots, e_n\}$. Thus, for $x \in \mathbb{R}^n$:

$$x = \sum_{i=1}^n x_i e_i$$

Thus by properties of norm:

$$\|x\| = \left\| \sum_{i=1}^n x_i e_i \right\| \leq \sum_{i=1}^n |x_i| \|e_i\|$$

By Cauchy-Schwarz inequality,

$$\sum_{i=1}^n |x_i| \|e_i\| \leq \left(\sum_{i=1}^n \|e_i\|^2 \right)^{1/2} \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2} = \left(\sum_{i=1}^n \|e_i\|^2 \right)^{1/2} \|x\|_{Euc}$$

Define:

$$B = \left(\sum_{i=1}^n \|e_i\|^2 \right)^{1/2}$$

Thus,

$$\|x\| \leq B \|x\|_{Euc}$$

Fix $\epsilon > 0$ pick $\delta = \epsilon / B$.

Let $x \in \mathbb{R}^n$:

If $\|x - y\|_{Euc} < \delta$ then by reverse triangle inequality

$$|\|x\| - \|y\|| \leq \|x - y\| \leq B \|x - y\|_{Euc} < \epsilon$$

Thus f is continuous.

- (b) $\|\cdot\|_{Euc}$ is standard norm.

Consider $f : S^{n-1} = \{x \in \mathbb{R}^n \mid \|x\|_{Euc} = 1\} \rightarrow S$ s.t.

$$f(x) = \frac{x}{\|x\|}$$

This is continuous since by (a) $\|x\|$ is continuous and since $x \neq 0$ for $x \in S^{n-1}$ thus

$\frac{x}{\|x\|}$ is continuous.

Since S^{n-1} is closed and bounded it is compact.

Consider $f(S^{n-1})$.

Let $x \in f(S^{n-1})$ then there is a $y \in S^{n-1}$ s.t. $f(y) = x \implies x = y/(\|y\|)$. Thus, $\|x\| = \|y/\|y\|\| = \|y\|/\|y\| = 1$. Thus, $x \in S$.

Let $x \in S$. Consider $y = x/\|x\|_{Euc}$

$$f(y) = y/\|y\| = \frac{x/\|x\|_{Euc}}{\|x/\|x\|_{Euc}\|} = x$$

Thus, $S = f(S^{n-1})$

Thus, S is compact since it is the image of a compact set from a continuous function.

(c) Consider the function $f : S \rightarrow \mathbb{R}$ by $f(x) = \|x\|_2$ where $S = \{x \in \mathbb{R}^n \mid \|x\|_1 = 1\}$.

By (a) we know that f is continuous, and since the domain of f is compact we know that f achieves a max and min on $\mathbb{R}$.

Let $\max = M$ and $\min = m$.

Thus, for all $x \in S$ we know that

$$m \leq \|x\|_2 \leq M$$

Let $x \in \mathbb{R}^n$.

$$\text{Consider } \|x\|_2 = \|x\|_1 \left\| \frac{x}{\|x\|_1} \right\|_2.$$

Since $x/\|x\|_1 \in S$

We get that

$$m \leq \|x/\|x\|_1\|_2 \leq M$$

$$m\|x\|_1 \leq \|x\|_2 \leq M\|x\|_1$$

Thus let $A = \max\{1/m, M\}$

$$\text{Thus } \frac{1}{A}\|x\|_1 \leq \|x\|_2 \leq A\|x\|_1$$

Let U be open in $(\mathbb{R}^n, \|\cdot\|_1)$.

Thus for $x \in U$ there is an $\epsilon > 0$ s.t. $B_1(x, \epsilon) = \{y \in \mathbb{R}^n \mid \|y - x\|_1 < \epsilon\} \subseteq U$.

$$\text{Let } y \in B_2(x, \epsilon/A) = \{y \in \mathbb{R}^n \mid \|y - x\|_2 < \epsilon/A\}$$

$$\|y - x\|_1 \leq A\|y - x\|_2 < \epsilon$$

Thus $y \in B_1(x, \epsilon)$

Therefore $B_2(x, \epsilon/A) \subseteq U$. Therefore, U is open in $(R^n, \|\cdot\|_2)$

Let U be open in $(R^n, \|\cdot\|_2)$

Therefore, for $x \in U$ there is an $\epsilon > 0$ s.t. $B_2(x, \epsilon) \subseteq U$.

Let $y \in B_1(x, \epsilon/A)$

$$\|y - x\|_2 \leq A\|y - x\|_1 < \epsilon$$

Thus $y \in B_1(x, \epsilon) \subseteq U$

Therefore, U is open in $(R^n, \|\cdot\|_1)$

Thus, these norms produce the same topology.